

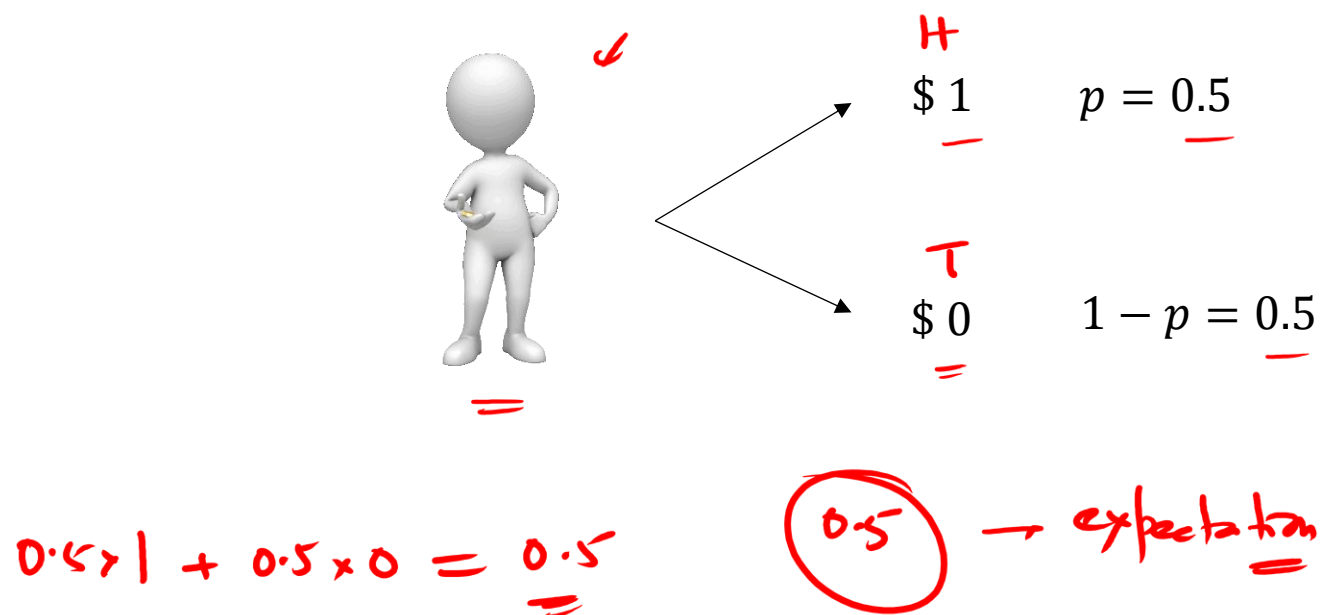
Derivative Pricing

(1. The Binomial Model)

I. Setting the Stage

Pricing a Gamble (I)

- Consider a coin toss game. You win \$1 if it shows head and nothing if it shows tail. How much will you pay for the ticket to play this game assuming the coin is fair ?



Pricing a Gamble (II)

- The simplest pricing strategy is **expectation**. Expectation price for the gamble is 0.5 \$. This appears to be the fair price of the ticket. In fact, **strong law of large numbers** states that sample mean of i.i.d random variables converges to population mean if sample size is large.

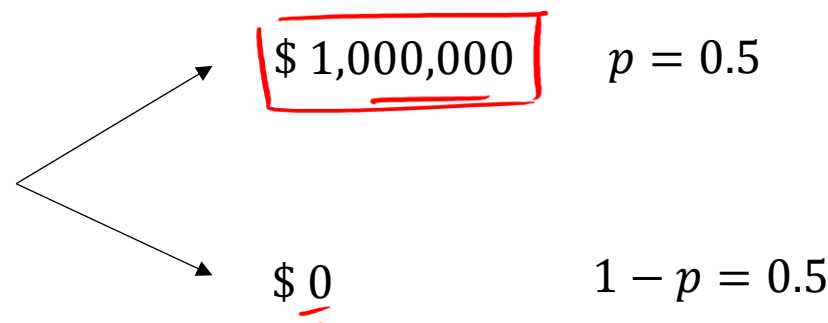
$n \times 0.5$ → $(S_n) = (X_1 + X_2 + \dots + X_n)/n$ approaches μ with probability 1 when $n \rightarrow \infty$

$\mu = E[X]$

So, if you keep playing this game over and over, in the long run your average winning will approach \$ 0.5. Now let's up the ante. Instead of \$1, you get \$1 million. How much will you pay now for the ticket ?

$E[X] = 500k$

$p < 500k$



Expectation Pricing

- Expectation pricing does not work in real life. This is because investors are in general risk averse. Risk erodes the utility. The price is typically lower than the raw expectation.

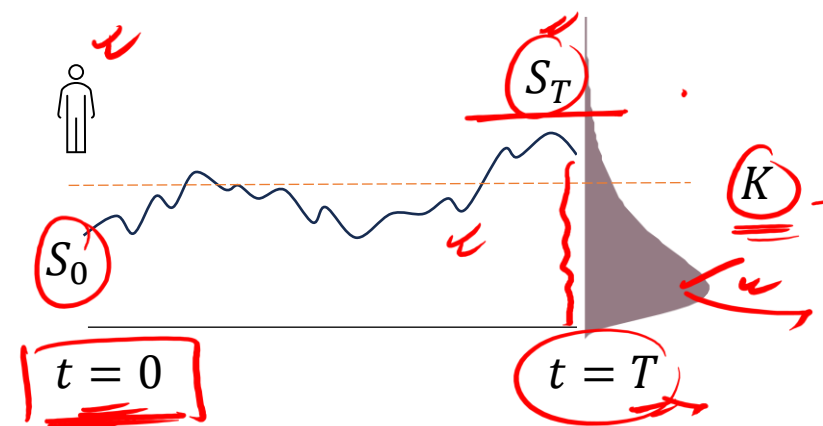
$$\underline{V} = \underline{E}(r) - \underbrace{\lambda}_{\substack{\uparrow \\ -ve}} \underbrace{(\sigma^2)}_{\substack{\downarrow \\ -ve}}$$

Expectation price is a good starting point, but it's not enforceable.

Pricing a Forward (I)

- An equity **forward contract** is an agreement to exchange a stock at a pre-determined price on a future date. This price is called the **strike price (K)** and is nothing but the forward price of the stock at time $t = 0$, i.e., $K = F(0, T)$. For the buyer of the contract (long position), the pay-off at maturity is given as

(forward payoff) $(V_T) = (S_T - K)$



S_T → random

- The strike price (K) is determined at inception such that the price of the forward = 0 at the inception. How to find K ?

Pricing a Forward (II)

- Common sense dictates $K = \mathbb{E}[S_T]$. Let's assume risk-free rate is constant (r). Let's assume the stock price in the real world has the following dynamics below.

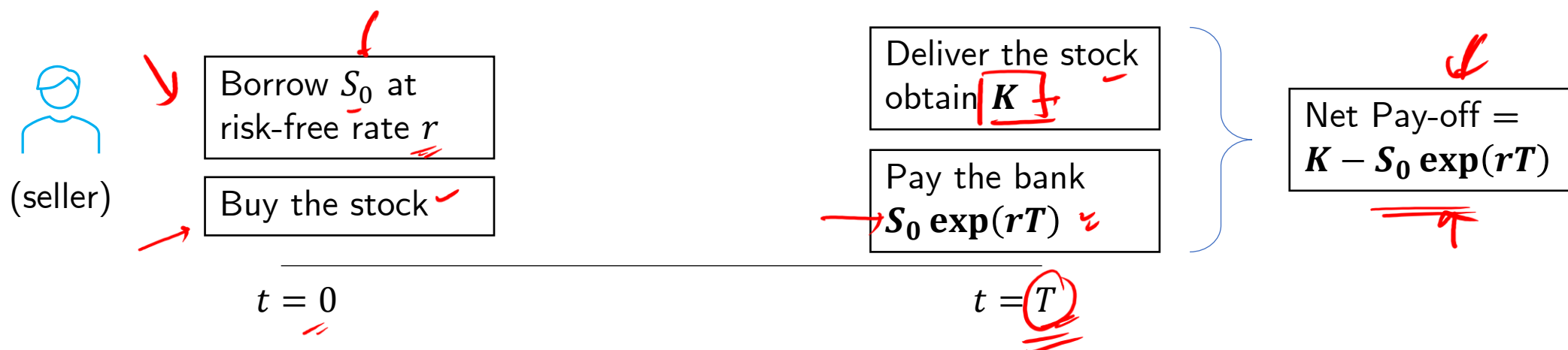
Stock price follows the process : $dS_t = \mu S_t dt + \sigma S_t dW_t$ (GBM)

Stock price at time T (lognormal) : $S_T = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right]$

We know that $\mathbb{E}[S_T]$ = $S_0 \exp(\mu T)$. Then, is $K = S_0 \exp(\mu T)$??

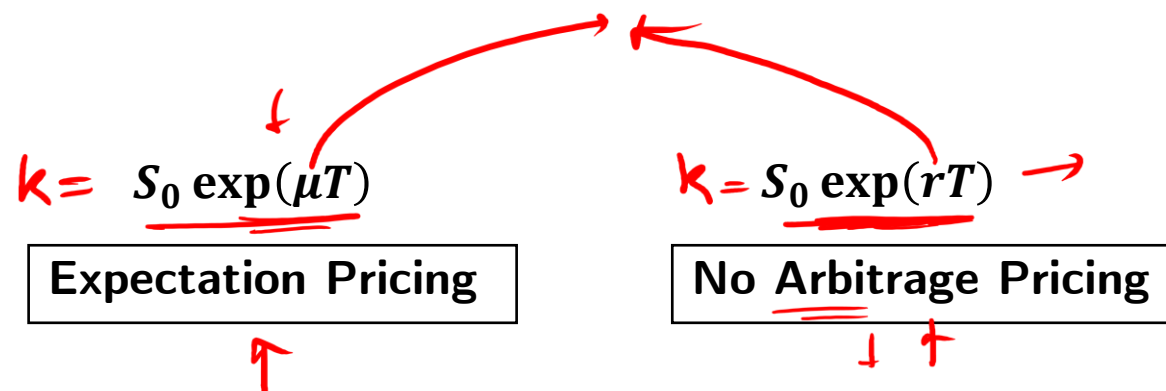
Pricing a Forward (III)

- ▶ Based on our experience with “pricing a gamble”, we know that expectation pricing is not enforceable. Then what is enforceable? Is there a stronger argument?
- ▶ Yes, there is. The single most important law in pricing is called “No Arbitrage Pricing”. This basically means there is no surefire way to make riskless profit.
- ▶ The Forward Seller has the obligation to deliver the stock at $t = T$ for a price K . He follows this strategy. At $t = 0$, borrow cash S_0 at risk-free rate r and buy the stock in the spot market. At $t = T$, deliver the stock and obtain a price K . Pay the bank an amount $S_0 \exp(rT)$. Since no money is spent at $t = 0$, no money should be gained with certainty at $t = T$. This gives $K = S_0 \exp(rT)$



Is Expectation Pricing worthless ?

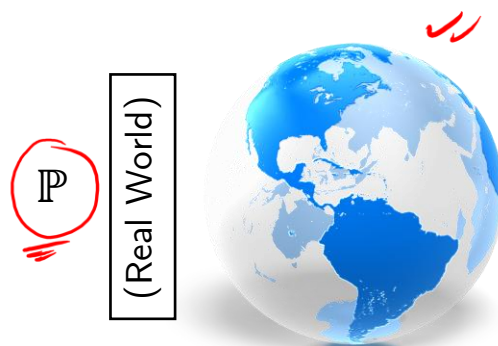
- We see that no arbitrage principle gives a different answer compared to expectation pricing. Does it mean Expectation pricing is dead ? Expectation is such a nice and easy tool, at least compared to the no arbitrage set up which involves very creative ideas to set up no arbitrage portfolio. But it gives an incorrect price.



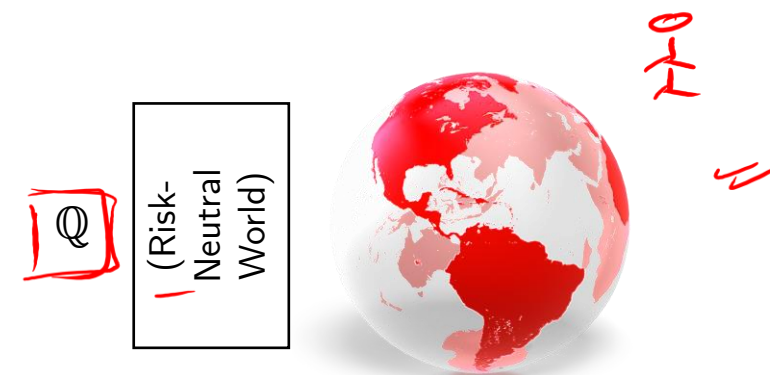
- Can we still use Expectation pricing somehow ? Life would be so simple!

Expectation Pricing – Recovered!

- **Yes, we can**, but we have to abandon this world. If you see the two expressions, the only difference is in the drift. So, if pretend the drift is r and not μ , we can still do Expectation based Pricing !! This new world is called **risk-neutral** world, where everything grows as risk-free rate r . From now on we will refer to real world as $\mathbb{P}$ measure and risk-neutral as $\mathbb{Q}$ measure



$$\begin{aligned} \rightarrow dS_t &= \underline{\mu} S_t dt + \sigma S_t dW_t \quad (\text{SDE}) \\ S_T &= S_0 \exp \left[\left(\underline{\mu} - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right] \quad \text{solution} \\ \rightarrow \mathbb{E}[S_T] &= S_0 \exp(\underline{\mu} T) \quad \text{expectation} \end{aligned}$$



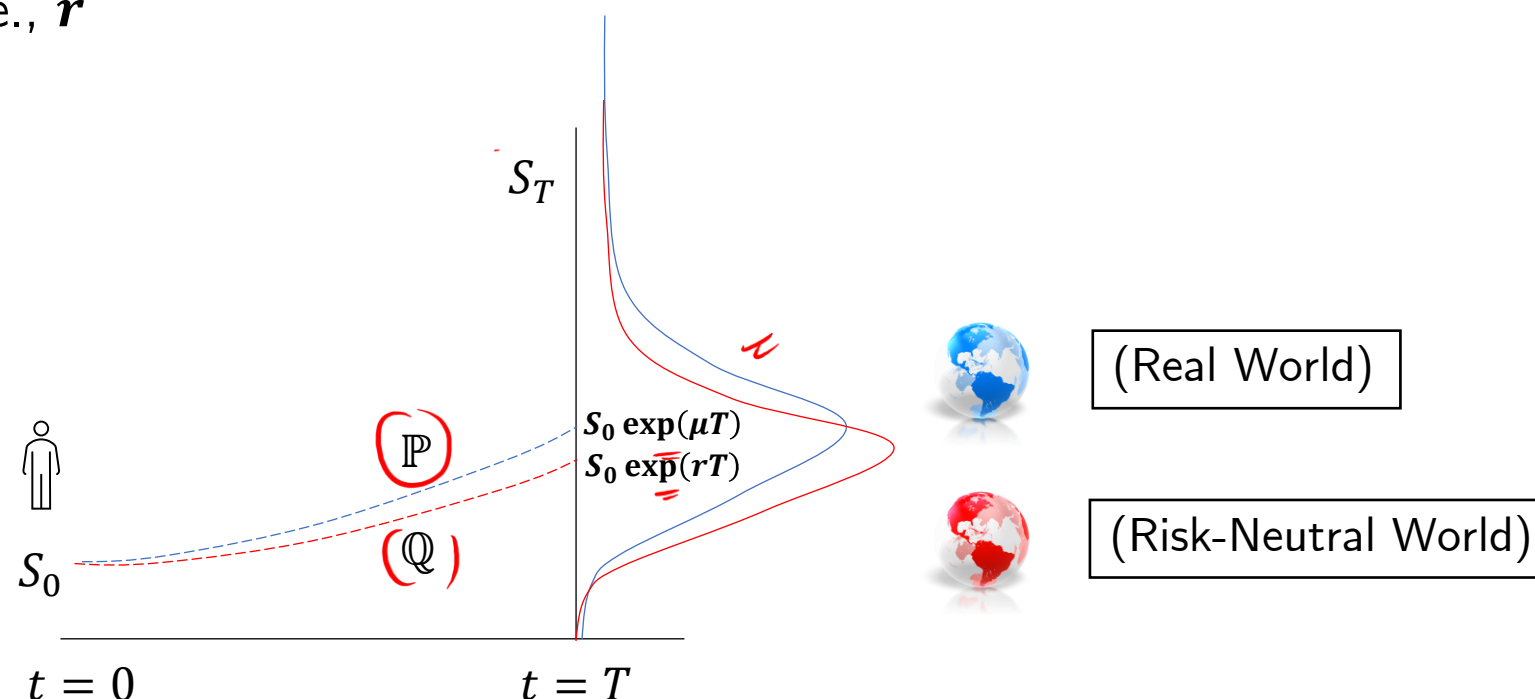
$$\begin{aligned} dS_t &= \underline{r} S_t dt + \sigma S_t dW_t \\ S_T &= S_0 \exp \left[\left(\underline{r} - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right] \\ \rightarrow \mathbb{E}[S_T] &= S_0 \exp(\underline{r} T) \quad \leftarrow \end{aligned}$$

Real World vs Risk Neutral

- The actual dynamics of the stock in the real world still has a drift of μ . Only because we wanted to use Expectation pricing to match no arbitrage pricing, we had to change the drift from μ to r . Note that only the drift is changed **not the volatility**. This new world where everything grows at risk-free rate is called risk-neutral because investors do not care about risk. They don't want compensation for investing in a risky asset. For them, the average growth rate for any risky asset should be the same as the constant growth rate i.e., r

$$\begin{aligned} \mu &> r \quad \text{--- (1)} \\ \mu &< r \quad \text{--- (2)} \end{aligned}$$

\$1m
\$0
< 500k



Exercise

Suppose S_t has the following SDE

$$\underline{dS_t/S_t} = \underline{\mu dt} + \sigma dW_t \quad \text{hansen} \quad \text{hansen} \quad \text{GPM}$$

Evaluate the integral $\int_0^\infty (S_T - K) f(S_T) dS_T$ where $f(S_T)$ is the density of S_T

$$\int x f(x) dx$$

$$\begin{aligned} \int_0^\infty (S_T - K) f(S_T) dS_T &= \int_0^\infty S_T f(S_T) dS_T - K \int_0^\infty f(S_T) dS_T \\ &= \mathbb{E}[S_T] - K \end{aligned}$$

The integral evaluates

E x p e c t e d

p a y o f f

$$= S_0 e^{rT} - K$$

$$, \underline{S_0 e^{rT} - K}$$

$$V_0 = (S_0 e^{rT} - K) e^{-rT} = S_0 - \underline{K e^{-rT}} = (0) \leftarrow$$

II. Discrete Pricing Model



Introduction to discrete model

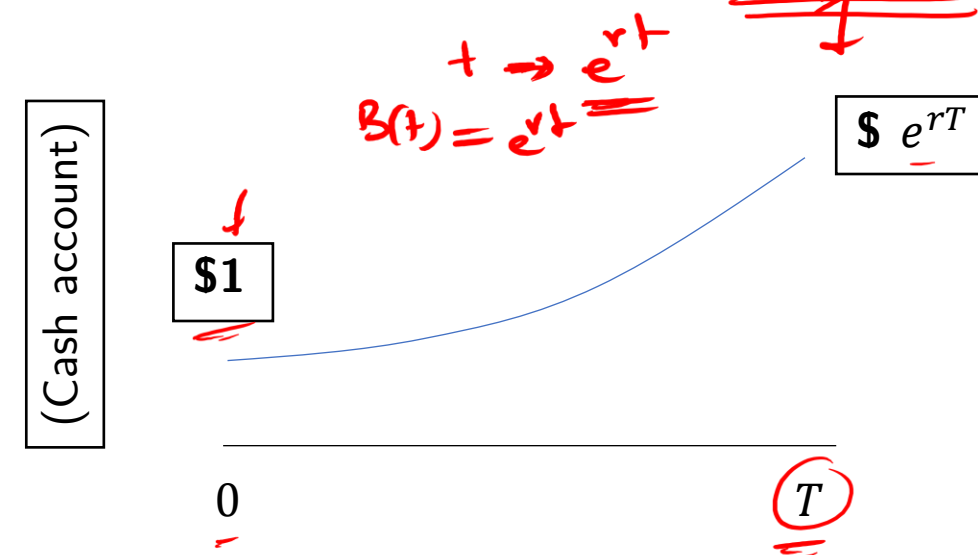
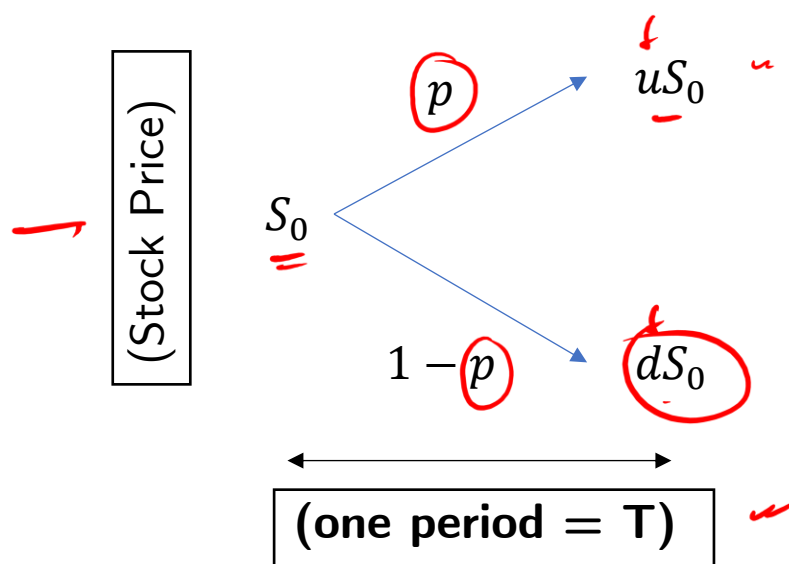
- We will start with a discrete model for stock price. The model is naïve but has some powerful ideas that emerge clearly out of it such as **Martingales**, **Risk-neutral** pricing. Also, the discrete model gives prices for American options as well. It's also possible to derive **Black-Scholes** equation from the discrete model in the limiting case.

Assumptions →

- a) No Trading costs, taxes
- b) No minimum or maximum units of trading
- c) Stocks and bonds can be bought and sold at discrete times $t=1,2,\dots$
- d) Principle of no-arbitrage applies

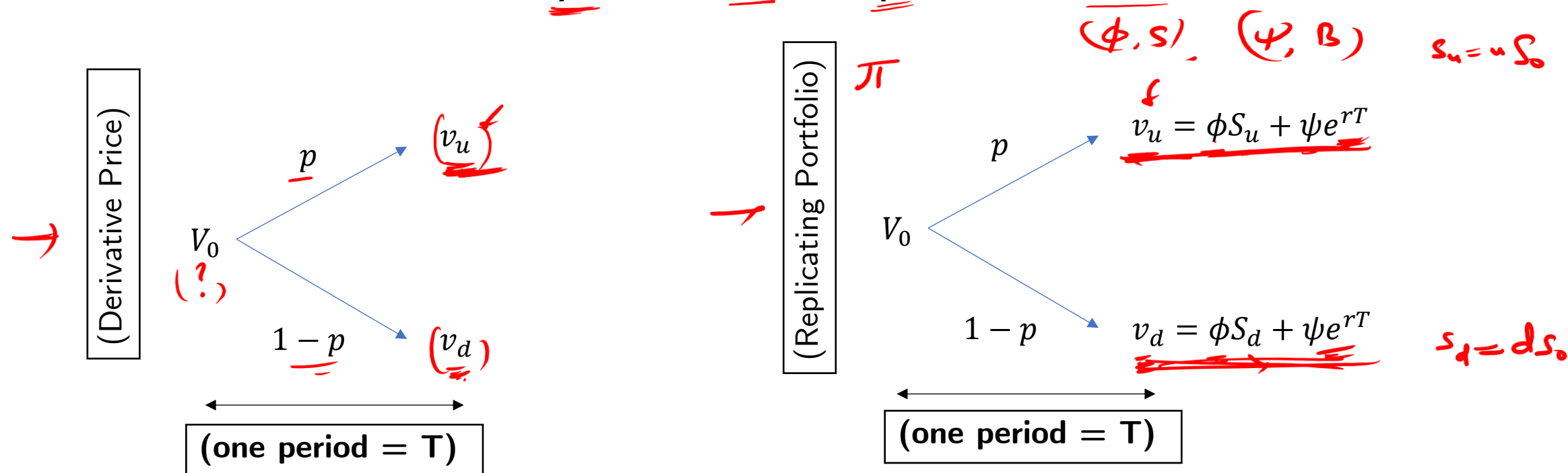
Dynamics of Stock and Bank Account

- ▶ We will assume the following dynamics for the stock price and the bank account. In one period (T), the stock price can go up or down by a factor u and d respectively with real-world probabilities p and $1 - p$, respectively. The stock price does not pay any dividends. This is called the **Binomial Model**. Each path is a **branch**, and the end of each branch is called a **node**.
- ▶ For the bank account B , we will assume it grows continuously with risk-free rate r which is constant. Assume without loss of generality, the initial value of the bank account, $B_0 = 1$. At any time t , in future, the value of this account is given with certainty e^{rt} . We must have $d < e^{rT} < u$



Pricing a Derivative (I)

- ▶ Consider a derivative which pays v_u when the stock goes up and v_d when the stock goes down. What is the price of the derivative now $V_0 = ?$
- ▶ We will use a synthetic portfolio π of the stock and the cash account such that the payoff of the synthetic portfolio at the end of the period matches exactly the derivative price irrespective of up state or down state. Assume we hold ϕ units of stock and ψ units of bank account.



Pricing a Derivative (II)

- This gives us 2 equations and 2 unknowns which is easily solved

$$\begin{aligned} \underline{v_u} &= \underline{\phi S_u} + \underline{\psi e^{rT}} \\ \underline{v_d} &= \underline{\phi S_d} + \underline{\psi e^{rT}} \end{aligned}, \quad \begin{aligned} S_u &= uS_0 \\ S_d &= dS_0 \end{aligned}$$

$$\underline{\phi = \frac{v_u - v_d}{S_0(u - d)}} \quad \& \quad \underline{\underline{\psi = e^{-rT} \left(\frac{v_d u - v_u d}{u - d} \right)}}$$

- Since this portfolio $\pi(\phi, \psi)$ perfectly replicates the derivative's payoff, the price of this portfolio today must be the same as the derivative's price today. (No Arbitrage)

$$\underline{V_0} = \underline{\phi S_0} + \underline{\psi \times 1} = \underline{\phi S_0} + \underline{\psi}$$

Pricing a Derivative (III)

► Replacing the values of ϕ, ψ , we get

(ϕ, ψ)

$$V_0 = e^{-rT} \left[\left(\frac{e^{rT} - d}{u - d} \right) v_u + \left(1 - \frac{e^{rT} - d}{u - d} \right) v_d \right] \rightarrow$$

discounted expected payoff

which is $V_0 = e^{-rT} [(q)v_u + (1-q)v_d]$ where, $q = \frac{e^{rT} - d}{u - d}$

$d < e^{rT} < u$

< 1

The price today appears as the one period discounted expectation under a new probability measure $\mathbb{Q} (q, 1-q)$. Note that q qualifies to be called as a probability because $0 < q < 1$. We can denote the price today as

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[V_T]$$

Diagram illustrating the relationship between V_0 , V_T , and the discount factor e^{-rT} .

$V_T \rightarrow V_u$
 $V_T \rightarrow V_d$
 $V_0 = \mathbb{E}^{\mathbb{Q}} [e^{-rT} V_T]$

Exercise

1. Find the following

a) $\mathbb{E}^{\mathbb{P}}[S_T]$ →

b) $\mathbb{E}^{\mathbb{Q}}[S_T]$

→ And show that the stock price today is consistent with the risk-neutral framework.

2. Argue why would normally $p > q$ →

3. Assume one period binomial model with $T=1$. Current stock price is 40. Stock price can move up to 60 or move down to 30 with probabilities 0.5 each. The continuously compounded risk-free rate = 5% per period. Find the current value of a European call option with strike price 45 using (S.) (r.)

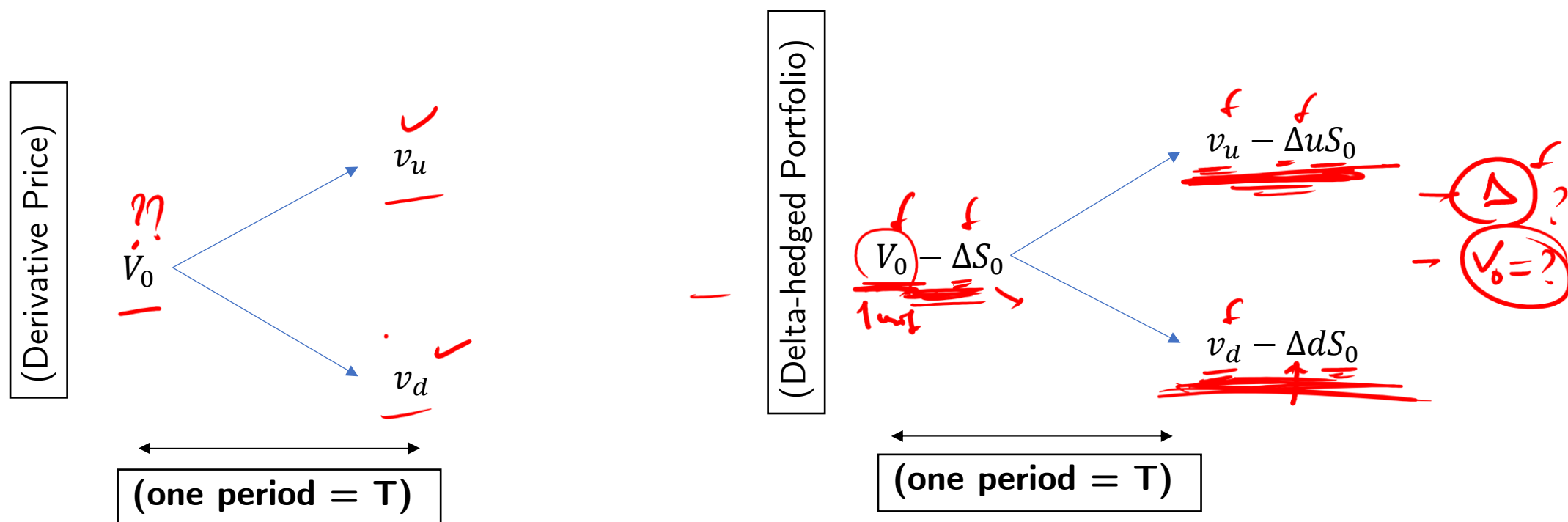
a) Replicating Portfolio strategy ✓

b) Risk-Neutral expectation strategy ✓

4. Derive the forward price for the stock using risk-neutral measure.

Delta-Hedge strategy (I)

- We can also impose the no-arbitrage pricing in a slightly different way. Consider a portfolio consisting of (long **1 unit** of derivative and short **Δ unit** of stock). We choose **Δ** in such a way that the portfolio is delta-hedged or risk-free. Then, to prevent arbitrage this portfolio must also grow at risk-free rate same as the bank account.



Delta-Hedge strategy (II)

- Since this portfolio is risk-free, it must satisfy two conditions

(Payoff must be same in up state and down state)

$$\underline{v_u - \Delta u S_0} = \underline{v_d - \Delta d S_0}$$

(risk-free growth)

$$\underline{v_u - \Delta u S_0} = \underline{e^{rT} (V_0 - \Delta S_0)}$$

Solving the above two equations for (V_0, Δ), we obtain

$$\underline{\phi = (\Delta)} = \underline{\frac{v_u - v_d}{S_0(u - d)}} \quad \& \quad \underline{V_0} = e^{-rT} \left[\left(\frac{e^{rT} - d}{u - d} \right) v_u + \left(1 - \frac{e^{rT} - d}{u - d} \right) v_d \right]$$

This is the same result we obtained under replicating portfolio strategy.